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5XXX ALUMINUM ALLOYS AND METHODS OF MAKING THE SAME

Abstract

New 5xxx aluminum alloy sheet products and methods for making the same are disclosed. The new 5xxx aluminum alloy sheet products may realize a low yield point elongation (YPE) thereby facilitating reduction or elimination of Type A Lüdering (stretcher-strain lines). The new 5xxx aluminum alloy sheet products may be hot rolled into an intermediate gauge product, then cold rolled to a final gauge product, wherein no intermediate annealing occurs prior to or during the cold rolling, and then post cold roll annealing at a temperature and for a time sufficient to first realize a recovered but unrecrystallized microstructure, and then converting of at least some of the recovered but unrecrystallized microstructure to a recrystallized microstructure via a second annealing operation of the post cold roll annealing step.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] This application is a continuation of International Patent Application No. PCT/US2023/083455, entitled “New 5XXX Aluminum Alloys and Methods of Making the Same,” filed Dec. 11, 2023, which claims the benefit of U.S. Provisional Patent Application No. 63/432,022, entitled “NEW 5XXX ALUMINUM ALLOYS AND METHODS OF MAKING THE SAME,” filed Dec. 12, 2022. Each of the above-identified patent applications is hereby incorporated by reference in its entirety.

BACKGROUND

[0002] 5xxx aluminum alloys are aluminum alloys having magnesium as the primary alloying element besides aluminum. Improving one or more properties of an aluminum alloy without degrading other properties is elusive. For instance, in automotive applications, it can be important to avoid Type A Lüdering (stretcher-strain lines) while also realizing high strength.

SUMMARY OF THE DISCLOSURE

[0003] Broadly, the present patent application relates to new methods of processing 5xxx aluminum alloys. The new methods may, for instance, facilitate realization of aluminum alloy sheets having high strength while also restricting or preventing Type A Lüdering. The new methods (processing) and associated compositions, microstructures, properties, and product applications are described below. Definitions are also provided below.

i. Processing

[0004] Referring now to FIG. 1, one non-limiting embodiment of a new method for producing 5xxx aluminum alloys is illustrated. In the illustrated embodiment, the method (**10**) includes hot rolling a 5xxx aluminum alloy into an intermediate gauge product (**100**), and then conducting a cold rolling operation to achieve a final gauge product (**200**), wherein the cold rolling operation includes at least one cold rolling step, and wherein no intermediate annealing (**250**) occurs prior to or during the cold rolling operation, and then post cold roll annealing the final gauge product (**300**). The post cold roll annealing (**300**) may be conducted at one or more temperatures and for one or more time periods. In one embodiment, the post cold roll annealing includes a first annealing step wherein, due to the first annealing step, the final gauge product realizes a recovered but unrecrystallized microstructure. A second annealing step may be conducted after the first annealing step, wherein the first annealing step converts at least some of the recovered but unrecrystallized microstructure to a recrystallized microstructure (defined below). In one embodiment, the post cold roll annealing step may be conducted at one or more temperatures and for one or more time periods sufficient for the final gauge product to realize: (i) a YPE of not greater than 0.60%, (ii) no Type A Lüdering, or (iii) both (i) and (ii).

A. Hot Rolling

[0005] The hot rolling step (**100**) generally comprises hot rolling an ingot or a continuously cast strip of a 5xxx aluminum alloy into the intermediate gauge product. The ingot may be produced by conventional direct chill casting operations. The continuously cast strip may be produced by conventional strip casting operations. In either case, the ingot or continuously cast strip may be homogenized prior to the hot rolling step (**100**). Any number of hot rolling passes may be used to produce an intermediate gauge product. In one embodiment, the intermediate gauge product has a thickness sufficient to yield a final gauge product of from 0.5 to 5.0 mm after completion of the cold rolling operation (**200**), as described in further detail below.

B. Cold Rolling Operation

[0006] After hot rolling (**100**), the cold rolling operation (**200**) is completed. The cold rolling operation (**200**) may include any number of cold rolling steps. The cold rolling operation (**200**) does not include an intermediate annealing steps (**250**), thereby saving time and allowing for more efficient processing.

[0007] In one embodiment, the cold rolling operation is accomplished with a single cold rolling pass. Unexpectedly, it has been found that due to the unique processing steps described herein, the final gauge product may realize a YPE of not greater than 0.60% despite a large amount of cold work being applied, such as in a single cold rolling pass and without an intermediate annealing. Nonetheless, the cold rolling operation (**200**) may be accomplished with one or multiple cold rolling passes.

[0008] In one embodiment, the cold rolling operation (**200**) comprises reducing a thickness of the intermediate gauge product by at least 25%. In another embodiment, the cold rolling operation (**200**) comprises reducing a thickness of the intermediate gauge product by at least 30%. In yet another embodiment, the cold rolling operation (**200**) comprises reducing a thickness of the intermediate gauge product by at least 35%. In another embodiment, the cold rolling operation (**200**) comprises reducing a thickness of the intermediate gauge product by at least 40%. In yet another embodiment, the cold rolling operation (**200**) comprises reducing a thickness of the intermediate gauge product by at least 45%. In another embodiment, the cold rolling operation (**200**) comprises reducing a thickness of the intermediate gauge product by at least 50%. In yet another embodiment, the cold rolling operation (**200**) comprises reducing a thickness of the intermediate gauge product by at least 55%. In another embodiment, the cold rolling operation (**200**) comprises reducing a thickness of the intermediate gauge product by at least 60%. In yet another embodiment, the cold rolling operation (**200**) comprises reducing a thickness of the intermediate gauge product by at least 65%.

[0009] In one embodiment, the cold rolling operation (**200**) comprises reducing the thickness of the intermediate gauge product by not greater than 80%. In another embodiment, the cold rolling operation (**200**) comprises reducing the thickness of the intermediate gauge product by not greater than 75%. In yet another embodiment, the cold rolling operation (**200**) comprises reducing the thickness of the intermediate gauge product by not greater than 70%.

[0010] In one approach, the final gauge product produced by the cold rolling operation (**200**) has a thickness of from 0.5 to 5.0 mm. In one embodiment, the final gauge product has a thickness of at least 1.0 mm. In one embodiment, the final gauge product has a thickness of not greater than 4.5 mm. In another embodiment, the final gauge product has a thickness of not greater than 4.0 mm. In yet another embodiment, the final gauge product has a thickness of not greater than 3.5 mm. In another embodiment, the final gauge product has a thickness of not greater than 3.0 mm. In yet another embodiment, the final gauge product has a thickness of not greater than 2.5 mm. In another embodiment, the final gauge product has a thickness of not greater than 2.4 mm. In yet another embodiment, the final gauge product has a thickness of not greater than 2.3 mm. In another embodiment, the final gauge product has a thickness of not greater than 2.2 mm. In yet another embodiment, the final gauge product has a thickness of not greater than 2.1 mm. In another embodiment, the final gauge product has a thickness of not greater than 2.0 mm.

C. Post Cold Roll Anneal

[0011] As noted above, after the cold rolling operation (**200**), the method (**10**) comprises a post cold rolling anneal step (**300**). Referring now to FIG. 2, in one embodiment, the post cold rolling anneal step (**300**) comprises first annealing at a first temperature for a first duration (**320**), and second annealing at a second temperature for a second duration (**340**). Due to the first annealing step (**320**), the final gauge product may realize a recovered but unrecrystallized microstructure (**322**). Due to the second annealing step (**340**), the final gauge product may realize a recrystallized microstructure (**342**). In one embodiment, the second annealing step (**340**) may convert at least

some of the recovered but unrecrystallized microstructure (322) to a recrystallized microstructure (342).

[0012] The temperature of the first annealing step (320) may include one temperature set-point or may include multiple temperature set-points. The duration of the first annealing step (320) may include one duration or multiple durations and across one or more temperatures of the first annealing step (320). The temperature of the second annealing step (340) may include one temperature set-point or may include multiple temperature set-points. The duration of the second annealing step (340) may include one duration or multiple durations and across one or more temperatures of the second annealing step (340). In one embodiment, the temperature of the first annealing step (320) is below a recrystallization temperature of the final gauge product, and the temperature of the second annealing step (340) is above a recrystallization temperature of the final gauge product.

[0013] In approach embodiment, the temperature of the first annealing step (320) (“the first temperature”) is from 350° F. to 500° F. In one embodiment, the first temperature is at least 375° F. In another embodiment, the first temperature is at least 400° F. In yet another embodiment, the first temperature is at least 410° F. In another embodiment, the first temperature is at least 420° F. In yet another embodiment, the first temperature is at least 430° F. In another embodiment, the first temperature is at least 440° F. In yet another embodiment, the first temperature is at least 450° F. In one embodiment, the first temperature is not greater than 490° F. In another embodiment, the first temperature is not greater than 480° F. In yet another embodiment, the first temperature is not greater than 470° F. In another embodiment, the first temperature is not greater than 460° F.

[0014] The duration of the first annealing step (320) (“the first duration”) is generally at least 30 minutes. In one approach, the first duration is from at least 1 hour to not greater than 10 days. In one embodiment, the first duration is at least 2 hours. In another embodiment, the first duration is at least 4 hours. In yet another embodiment, the first duration is at least 6 hours. In another embodiment, the first duration is at least 8 hours. In yet another embodiment, the first duration is at least 10 hours. In another embodiment, the first duration is at least 12 hours. In yet another embodiment, the first duration is at least 14 hours. In another embodiment, the first duration is at least 16 hours. In yet another embodiment, the first duration is at least 18 hours. In another embodiment, the first duration is at least 20 hours. In yet another embodiment, the first duration is at least 22 hours. In another embodiment, the first duration is at least 24 hours. In yet another embodiment, the first duration is at least 26 hours. In another embodiment, the first duration is at least 28 hours. In yet another embodiment, the first duration is at least 30 hours. In another embodiment, the first duration is at least 32 hours. In yet another embodiment, the first duration is at least 34 hours. In another embodiment, the first duration is at least 36 hours. In yet another embodiment, the first duration is at least 38 hours. In another embodiment, the first duration is at least 40 hours. In yet another embodiment, the first duration is at least 42 hours. In another embodiment, the first duration is at least 44 hours. In yet another embodiment, the first duration is at least 46 hours. In another embodiment, the first duration is at least 48 hours. In yet another embodiment, the first duration is at least 50 hours.

[0015] In one embodiment, the first duration is not greater than 9 days. In another embodiment, the first duration of the first annealing step is not greater than 8 days. In yet another embodiment, the first duration of the first annealing step is not greater than 7 days. In another embodiment, the first duration of the first annealing step is not greater than 6 days. In yet another embodiment, the first duration of the first annealing step is not greater than 5 days. In another embodiment, the first duration is not greater than 4 days. In yet another embodiment, the first duration is not greater than 3 days. In another embodiment, the first duration is not greater than 65 hours. In yet another embodiment, the first duration is not greater than 60 hours.

[0016] Due to the first temperature and first duration of the first annealing step (320) the final gauge 5xxx aluminum alloy product may realize a recovered but unrecrystallized microstructure

(322). In one embodiment, the combination of the first temperature and first duration of the first annealing step (320) are sufficient to achieve an unrecrystallized and recovered microstructure (322) in the final gauge 5xxx aluminum alloy product.

[0017] In one approach, the temperature of the second annealing step (340) (“the second temperature”) is at least 25° F. higher than the first temperature of the first annealing step (320). In one embodiment, the second temperature of the second annealing step (340) is at least 50° F. higher than the first temperature of the first annealing step (320). In another embodiment, the second temperature of the second annealing step (340) is at least 100° F. higher than the first temperature of the first annealing step (320). In yet another embodiment, the second temperature of the second annealing step (340) is at least 150° F. higher than the first temperature of the first annealing step (320). In another embodiment, the second temperature of the second annealing step (340) is at least 200° F. higher than the first temperature of the first annealing step (320). In yet another embodiment, the second temperature of the second annealing step (340) is at least 250° F. higher than the first temperature of the first annealing step (320). In another embodiment, the second temperature of the second annealing step (340) is at least 300° F. higher than the first temperature of the first annealing step (320). In yet another embodiment, the second temperature of the second annealing step (340) is at least 350° F. higher than the first temperature of the first annealing step (320). In another embodiment, the second temperature of the second annealing step (340) is at least 400° F. higher than the first temperature of the first annealing step (320).

[0018] In one approach, the second temperature of the second annealing step (340) is from more than 500° F. to 900° F. In one embodiment, the second temperature is at least 550° F. In another embodiment, the second temperature is at least 600° F. In yet another embodiment, the second temperature is at least 650° F. In another embodiment, the second temperature is at least 700° F. In yet another embodiment, the second temperature is at least 725° F. In another embodiment, the second temperature is at least 750° F. In yet another embodiment, the second temperature is at least 775° F. In another embodiment, the second temperature is at least 800° F. In yet another embodiment, the second temperature is at least 825° F. In one embodiment, the second temperature is not greater than 890° F. In another embodiment, the second temperature is not greater than 880° F. In yet another embodiment, the second temperature is not greater than 870° F. In another embodiment, the second temperature is not greater than 860° F. In yet another embodiment, the second temperature is not greater than 850° F. In another embodiment, the second temperature is not greater than 840° F.

[0019] Generally, the duration of the second annealing step (340) (“the second duration”) is at least 1 second (e.g., for a continuous in-line operation). In one approach, the second duration is from 5 seconds to 48 hours. In one embodiment, the second duration is at least 10 seconds. In another embodiment, the second duration is at least 15 seconds. In yet another embodiment, the second duration is at least 20 seconds. In another embodiment, the second duration is at least 30 seconds. In yet another embodiment, the second duration is at least 45 seconds. In another embodiment, the second duration is at least 60 seconds. In yet another embodiment, the second duration is at least 3 minutes (e.g., in batch anneal embodiments). In another embodiment, the second duration is at 5 minutes. In yet another embodiment, the second duration is at least 10 minutes. In another embodiment, the second duration is at 20 minutes. In yet another embodiment, the second duration is at least 30 minutes. In another embodiment, the second duration is at 45 minutes. In yet another embodiment, the second duration is at least 60 minutes. In another embodiment, the second duration is at 75 minutes. In yet another embodiment, the second duration is at least 90 minutes. In another embodiment, the second duration is at 105 minutes. In yet another embodiment, the second duration is at least 120 minutes.

[0020] In one embodiment, the second duration is not greater than 36 hours. In another embodiment, the second duration is not greater than 30 hours. In yet another embodiment, the second duration is not greater than 24 hours. In another embodiment, the second duration is not greater than 18 hours. In yet another embodiment, the second duration is not greater than 12 hours.

In another embodiment, the second duration is not greater than 11 hours. In yet another embodiment, the second duration is not greater than 10 hours. In another embodiment, the second duration is not greater than 9 hours. In yet another embodiment, the second duration is not greater than 8 hours. In another embodiment, the second duration is not greater than 7 hours. In yet another embodiment, the second duration is not greater than 6 hours. In another embodiment, the second duration is not greater than 5 hours. In yet another embodiment, the second duration is not greater than 4 hours.

[0021] Due to the second temperature and second duration of the second annealing step (340) the final gauge 5xxx aluminum alloy product may realize a recrystallized microstructure (342). In one embodiment, the combination of the second temperature and second duration of the second annealing step (320) are sufficient to achieve the recrystallized microstructure (342) in the final gauge 5xxx aluminum alloy product.

ii Composition

[0022] Any suitable 5xxx aluminum alloys may be processed in accordance with the new processing techniques disclosed herein. The new processing techniques may be especially useful in restricting or preventing Type A Lüdering in 5xxx aluminum alloys having high amounts of magnesium (e.g., from 4.0 to 6.0 wt. % Mg).

[0023] In one embodiment, the 5xxx aluminum alloy includes at least 4.0 wt. % Mg. In another embodiment, the 5xxx aluminum alloy includes at least 4.1 wt. % Mg. In yet another embodiment, the 5xxx aluminum alloy includes at least 4.2 wt. % Mg. In another embodiment, the 5xxx aluminum alloy includes at least 4.3 wt. % Mg. In yet another embodiment, the 5xxx aluminum alloy includes at least 4.4 wt. % Mg. In another embodiment, the 5xxx aluminum alloy includes at least 4.5 wt. % Mg.

[0024] In one embodiment, the 5xxx aluminum alloy includes not greater than 5.5 wt. % Mg. In another embodiment, the 5xxx aluminum alloy includes not greater than 5.4 wt. % Mg. In yet another embodiment, the 5xxx aluminum alloy includes not greater than 5.3 wt. % Mg. In another embodiment, the 5xxx aluminum alloy includes not greater than 5.2 wt. % Mg. In yet another embodiment, the 5xxx aluminum alloy includes not greater than 5.1 wt. % Mg. In another embodiment, the 5xxx aluminum alloy includes not greater than 5.0 wt. % Mg. In yet another embodiment, the 5xxx aluminum alloy includes not greater than 4.9 wt. % Mg. In another embodiment, the 5xxx aluminum alloy includes not greater than 4.8 wt. % Mg.

[0025] In one approach, a 5xxx aluminum alloy comprises from 4.0 to 5.0 wt. % Mg, up to 0.50 wt. % Mn, up to 0.75 wt. % Zn, up to 0.35 wt. % Fe, up to 0.20 wt. % Si, up to 0.15 wt. % Cu, up to 0.10 wt. % Cr, and up to 0.10 wt. % Ti, the balance being aluminum, optional incidental elements and impurities. In one approach, this 5xxx aluminum alloy includes less than 0.25 wt. % Zn, such as not greater than 0.20 wt. % Zn, or not greater than 0.15 wt. % Zn, or not greater than 0.10 wt. % Zn. In another approach this 5xxx aluminum alloy includes at least 0.25 wt. % Zn, such as or at least 0.30 wt. % Zn, or at least 0.35 wt. % Zn, or at least 0.40 wt. % Zn, or at least 0.45 wt. % Zn, or at least 0.50 wt. % Zn.

[0026] In one embodiment, the 5xxx aluminum alloy is selected from the group consisting of 5182, 5019, 5019A, 5119, 5119A, 5022, 5023, 5024, 5026, 5027, 5028, 5056, 5356, 5356A, 5456, 5456A, 5456B, 5556, 5556A, 5556B, 5556C, 5058, 5059, 5070, 5180, 5180A, 5082, 5083, 5183, 5183A, 5283, 5283A, 5283B, 5383, 5483, 5086, 5186, 5087, 5187, and 5088 aluminum alloys, as defined in the Aluminum Association document “*International Alloy Designations and Chemical Composition Limits for Wrought Aluminum and Wrought Aluminum Alloys*,” January 2015, The Aluminum Association Inc., 1525 Wilson Boulevard, Suite 600, Arlington, VA 22209. In one embodiment, the 5xxx aluminum alloy is 5082 or 5182. In one embodiment, the 5xxx aluminum alloy comprises from 4.0 to 5.0 wt. % Mg and is selected from the group consisting of 5182, 5019, 5019A, 5119, 5119A, 5022, 5024, 5026, 5027, 5028, 5056, 5356, 5356A, 5456, 5456A, 5456B, 5556, 5556C, 5058, 5070, 5180, 5180A, 5082, 5083, 5183, 5183A, 5283, 5283A, 5283B, 5383,

5483, 5086, 5186, 5087, 5187, and 5088 alloys.

[0027] As noted above, the balance of the 5xxx aluminum alloys is generally aluminum, optional incidental elements and impurities. As used herein, “incidental elements” means those elements or materials, other than the above listed elements, that may optionally be added to the alloy to assist in the production of the alloy. Examples of incidental elements include casting aids, such as grain refiners and deoxidizers. Optional incidental elements may be included in the alloy in a cumulative amount of up to 1.0 wt. %. As one non-limiting example, one or more incidental elements may be added to the alloy during casting to reduce or restrict (and in some instances eliminate) ingot cracking due to, for example, oxide fold, pit and oxide patches. These types of incidental elements are generally referred to herein as deoxidizers. Examples of some deoxidizers include Ca, Sr, and Be. When calcium (Ca) is included in the alloy, it is generally present in an amount of up to about 0.05 wt. %, or up to about 0.03 wt. %. In some embodiments, Ca is included in the alloy in an amount of about 0.001-0.03 wt. %, such as 0.001-0.008 wt. % (or 10 to 80 ppm). Strontium (Sr) may be included in the alloy as a substitute for Ca (in whole or in part), and thus may be included in the alloy in the same or similar amounts as Ca. Traditionally, beryllium (Be) additions have helped to reduce the tendency of ingot cracking, though for environmental, health and safety reasons, some embodiments of the alloy are substantially Be-free. When Be is included in the alloy, it is generally present in an amount of up to about 20 ppm. Incidental elements may be present in minor amounts, or may be present in significant amounts, and may add desirable or other characteristics on their own without departing from the alloy described herein, so long as the alloy retains the desirable characteristics described herein. It is to be understood, however, that the scope of this disclosure should not/cannot be avoided through the mere addition of an element or elements in quantities that would not otherwise impact on the combinations of properties desired and attained herein.

[0028] The 5xxx aluminum alloys may contain low amounts of impurities. In one embodiment, a 5xxx aluminum alloy includes not greater than 0.15 wt. %, in total, of the impurities, and wherein the 5xxx aluminum alloy includes not greater than 0.05 wt. % of each of the impurities. In another embodiment, a 5xxx aluminum alloy includes not greater than 0.10 wt. %, in total, of the impurities, and wherein the 5xxx aluminum alloy includes not greater than 0.03 wt. % of each of the impurities.

iii. Microstructure

[0029] The new 5xxx aluminum alloys described herein may realize a unique microstructure. In one embodiment, at least partially due to the combination of the first annealing step (320) and the second annealing step (340), the new 5xxx aluminum alloy product is recrystallized (342) as determined in accordance with the Microstructure Assessment Procedure, described in the Definitions section, below. In one embodiment, a new 5xxx aluminum alloy product is at least 60% recrystallized. In another embodiment, a new 5xxx aluminum alloy product is least 70% recrystallized. In yet another embodiment, a new 5xxx aluminum alloy product is least 75% recrystallized. In another embodiment, a new 5xxx aluminum alloy product is least 80% recrystallized. In yet another embodiment, a new 5xxx aluminum alloy product is least 85% recrystallized. In another embodiment, a new 5xxx aluminum alloy product is least 90% recrystallized. In yet another embodiment, a new 5xxx aluminum alloy product is least 95% recrystallized.

[0030] In one embodiment, at least partially due to the combination of the first annealing step (320) and the second annealing step (340), the final gauge product realizes at least one of (A) an area weighted average grain size of not greater than 40 micrometers (e.g., from 20-40 micrometers), (B) a Brass texture amount of not greater than 14%, and (C) an S texture amount of not greater than 19%, such as any of the average grain size, Brass texture quantities, and S texture quantities described below, and as determined in accordance with the Microstructure Assessment Procedure, described in the Definitions section, below. In another embodiment, at least partially due to the

combination of the first annealing step (320) and the second annealing step (340), the final gauge product realizes at least two of the (A), (B) and (C) properties mentioned above. In yet another embodiment, at least partially due to the combination of the first annealing step (320) and the second annealing step (340), the final gauge product realizes all three of the (A), (B) and (C) properties mentioned above.

[0031] As noted above, a new 5xxx aluminum alloy product may realize an area weighted average grain size of not greater than 40 micrometers. In one embodiment, a new 5xxx aluminum alloy product realizes an area weighted average grain size of at least 15 micrometers. In another embodiment, a new 5xxx aluminum alloy product realizes an area weighted average grain size of at least 18 micrometers. In yet another embodiment, a new 5xxx aluminum alloy product realizes an area weighted average grain size of at least 20 micrometers. In another embodiment, a new 5xxx aluminum alloy product realizes an area weighted average grain size of at least 22 micrometers. In yet another embodiment, a new 5xxx aluminum alloy product realizes an area weighted average grain size of at least 24 micrometers.

[0032] In one embodiment, a new 5xxx aluminum alloy product realizes an area weighted average grain size of not greater than 39 micrometers. In another embodiment, a new 5xxx aluminum alloy product realizes an area weighted average grain size of not greater than 38 micrometers. In another embodiment, a new 5xxx aluminum alloy product realizes an area weighted average grain size of not greater than 37 micrometers. In yet another embodiment, a new 5xxx aluminum alloy product realizes an area weighted average grain size of not greater than 36 micrometers.

[0033] In one embodiment, a new 5xxx aluminum alloy product realizes a Brass texture amount of not greater than 13%. In another embodiment, a new 5xxx aluminum alloy product realizes a Brass texture amount of not greater than 12%. In yet another embodiment, a new 5xxx aluminum alloy product realizes a Brass texture amount of not greater than 11%. In another embodiment, a new 5xxx aluminum alloy product realizes a Brass texture amount of not greater than 10%. In yet another embodiment, a new 5xxx aluminum alloy product realizes a Brass texture amount of not greater than 9%. In another embodiment, a new 5xxx aluminum alloy product realizes a Brass texture amount of not greater than 8%. In yet another embodiment, a new 5xxx aluminum alloy product realizes a Brass texture amount of not greater than 7%.

[0034] In one embodiment, a new 5xxx aluminum alloy product realizes a Brass texture amount of at least 1%. In another embodiment, a new 5xxx aluminum alloy product realizes a Brass texture amount of at least 2%. In another embodiment, a new 5xxx aluminum alloy product realizes a Brass texture amount of at least 3%. In another embodiment, a new 5xxx aluminum alloy product realizes a Brass texture amount of at least 4%. In another embodiment, a new 5xxx aluminum alloy product realizes a Brass texture amount of at least 5%. In another embodiment, a new 5xxx aluminum alloy product realizes a Brass texture amount of at least 6%.

[0035] In one embodiment, a new 5xxx aluminum alloy product realizes an S texture amount of not greater than 18%. In another embodiment, a new 5xxx aluminum alloy product realizes an S texture amount of not greater than 17%. In yet another embodiment, a new 5xxx aluminum alloy product realizes an S texture amount of not greater than 16%. In another embodiment, a new 5xxx aluminum alloy product realizes an S texture amount of not greater than 15%. In yet another embodiment, a new 5xxx aluminum alloy product realizes an S texture amount of not greater than 14%.

[0036] In one embodiment, a new 5xxx aluminum alloy product realizes an S texture amount of at least 5%. In another embodiment, a new 5xxx aluminum alloy product realizes an S texture amount of at least 7%. In another embodiment, a new 5xxx aluminum alloy product realizes an S texture amount of at least 9%. In another embodiment, a new 5xxx aluminum alloy product realizes an S texture amount of at least 11%. In another embodiment, a new 5xxx aluminum alloy product realizes an S texture amount of at least 12%. In another embodiment, a new 5xxx aluminum alloy product realizes an S texture amount of at least 13%.

realizes an area weighted average grain size that is at least 8 micrometers lower than an area weighted average grain size of a 5182-OE21 product of equivalent gauge. In another embodiment, a new 5xxx aluminum alloy product realizes an area weighted average grain size that is at least 9 micrometers lower than an area weighted average grain size of a 5182-OE21 product of equivalent gauge. In yet another embodiment, a new 5xxx aluminum alloy product realizes an area weighted average grain size that is at least 10 micrometers lower than an area weighted average grain size of a 5182-OE21 product of equivalent gauge. In another embodiment, a new 5xxx aluminum alloy product realizes an area weighted average grain size that is at least 11 micrometers lower than an area weighted average grain size of a 5182-OE21 product of equivalent gauge. In yet another embodiment, a new 5xxx aluminum alloy product realizes an area weighted average grain size that is at least 12 micrometers lower than an area weighted average grain size of a 5182-OE21 product of equivalent gauge. In another embodiment, a new 5xxx aluminum alloy product realizes an area weighted average grain size that is at least 13 micrometers lower than an area weighted average grain size of a 5182-OE21 product of equivalent gauge. In yet another embodiment, a new 5xxx aluminum alloy product realizes an area weighted average grain size that is at least 14 micrometers lower than an area weighted average grain size of a 5182-OE21 product of equivalent gauge. In another embodiment, a new 5xxx aluminum alloy product realizes an area weighted average grain size that is at least 15 micrometers lower than an area weighted average grain size of a 5182-OE21 product of equivalent gauge. In yet another embodiment, a new 5xxx aluminum alloy product realizes an area weighted average grain size that is at least 16 micrometers lower than an area weighted average grain size of a 5182-OE21 product of equivalent gauge. In another embodiment, a new 5xxx aluminum alloy product realizes an area weighted average grain size that is at least 17 micrometers lower than an area weighted average grain size of a 5182-OE21 product of equivalent gauge.

[0042] For instance, if a new 5xxx aluminum alloy product realizes an area weighted average grain size of 35 micrometers and a 5182-OE21 product of equivalent gauge realizes an area weighted average grain size of 45 micrometers, the area weighted average grain size of the new 5xxx aluminum alloy product is 10 micrometers lower than the area weighted average grain size of the 5182-OE21 product of equivalent gauge.

iv. Properties

[0043] As noted above, the new 5xxx aluminum alloys described herein may realize an improved combination of properties. For instance, products made from the new 5xxx aluminum alloys may realize an improved combination of two or more of (a) the absence or restricted presence of Type A Lüdering (stretcher-strain lines), (b) strength, (c) yield point elongation, and (d) ductility, such as a combination of two or more of the properties described below.

[0044] In one embodiment, a new final gauge 5xxx aluminum alloy product is free of Type A Lüdering (stretcher-strain lines).

[0045] As noted above, a new 5xxx aluminum alloy may realize a yield point elongation of (YPE) of not greater than 0.60%, which may be an indicator of Type A Lüdering propensity. In one embodiment, a new 5xxx aluminum alloy realizes a YPE of not greater than 0.55%. In another embodiment, a new 5xxx aluminum alloy realizes a YPE of not greater than 0.50%. In yet another embodiment, a new 5xxx aluminum alloy realizes a YPE of not greater than 0.45%. In another embodiment, a new 5xxx aluminum alloy realizes a YPE of not greater than 0.40%. In yet another embodiment, a new 5xxx aluminum alloy realizes a YPE of not greater than 0.35%. In another embodiment, a new 5xxx aluminum alloy realizes a YPE of not greater than 0.30%. In yet another embodiment, a new 5xxx aluminum alloy realizes a YPE of not greater than 0.25%. In another embodiment, a new 5xxx aluminum alloy realizes a YPE of not greater than 0.20%.

[0046] In one embodiment, a new 5xxx aluminum alloy realizes a tensile yield strength (LT) of at least 100 MPa. In another embodiment, a new 5xxx aluminum alloy realizes a tensile yield strength (LT) of at least 105 MPa. In yet another embodiment, a new 5xxx aluminum alloy realizes a tensile

yield strength (LT) of at least 110 MPa. In another embodiment, a new 5xxx aluminum alloy realizes a tensile yield strength (LT) of at least 115 MPa. In yet another embodiment, a new 5xxx aluminum alloy realizes a tensile yield strength (LT) of at least 120 MPa. In another embodiment, a new 5xxx aluminum alloy realizes a tensile yield strength (LT) of at least 125 MPa.

[0047] In one embodiment, a new 5xxx aluminum alloy realizes an ultimate tensile strength (LT) of at least 260 MPa. In another embodiment, a new 5xxx aluminum alloy realizes an ultimate tensile strength (LT) of at least 265 MPa. In yet another embodiment, a new 5xxx aluminum alloy realizes an ultimate tensile strength (LT) of at least 270 MPa. In another embodiment, a new 5xxx aluminum alloy realizes an ultimate tensile strength (LT) of at least 275 MPa. In yet another embodiment, a new 5xxx aluminum alloy realizes an ultimate tensile strength (LT) of at least 280 MPa. In another embodiment, a new 5xxx aluminum alloy realizes an ultimate tensile strength (LT) of at least 285 MPa.

[0048] In one embodiment, a new 5xxx aluminum alloy realizes a uniform elongation (LT) of at least 15%. In another embodiment, a new 5xxx aluminum alloy realizes a uniform elongation (LT) of at least 16%. In yet another embodiment, a new 5xxx aluminum alloy realizes a uniform elongation (LT) of at least 17%. In another embodiment, a new 5xxx aluminum alloy realizes a uniform elongation (LT) of at least 18%. In yet another embodiment, a new 5xxx aluminum alloy realizes a uniform elongation (LT) of at least 19%. In another embodiment, a new 5xxx aluminum alloy realizes a uniform elongation (LT) of at least 20%. In yet another embodiment, a new 5xxx aluminum alloy realizes a uniform elongation (LT) of at least 21%.

V. Product Applications

[0049] The new aluminum alloy described herein may be used in a variety of applications, such as an automotive, consumer electronic, and electrical applications, as well as in beverage can stock, among others. For example, a new aluminum alloy may be formed into an automotive part. Non-limiting examples of automotive parts include automotive bodies and automotive panels. Non-limiting examples of automotive panels may be outer panels, inner panels for use in car doors, car hoods, or car trunks (deck lids), among others. One example of an automotive body product may be a structural component, which are commonly sheet metal components of a car body (e.g., body-in-white) where additional strength is required to withstand crash requirements. In one embodiment, the new aluminum alloy is an enclosure for a battery, such as a battery used in an electric vehicle. The new aluminum alloys may also be used in other transportation applications, such as light or heavy trucks. Consumer electronic product applications include laptop computer cases, battery cases, among other stamped and formed products. Beverage can stock includes can tab and can end applications.

vi. Definitions

[0050] “Wrought aluminum alloy product” means an aluminum alloy product that is hot worked after casting, and includes rolled products (sheet or plate), forged products, and extruded products.

[0051] “Hot working” such as by hot rolling means working the aluminum alloy product at elevated temperature, and generally at least 121.1° C. (250° F.). Strain-hardening is restricted/avoided during hot working, which generally differentiates hot working from cold working.

[0052] “Cold working” such as by cold rolling means working the aluminum alloy product at temperatures that are not considered hot working temperatures, generally below about 121.1° C. (250° F.) (e.g., at ambient).

[0053] Temper definitions are per ANSI H35.1 (2009), entitled “American National Standard Alloy and Temper Designation Systems for Aluminum,” published by The Aluminum Association.

[0054] Strength and elongation are measured in accordance with ASTM E8/E8M-16a and B557-15.

[0055] “Intermediate anneal” refers to a purposeful heat treatment conducted relative to a material while that material is at an intermediate gauge. For instance, an intermediate anneal may be an anneal conducted before or between cold rolling passes relative to an intermediate gauge material.

[0056] “Yield Point Elongation” (YPE) is to be determined as follows: [0057] First, a stress-strain curve of the aluminum alloy is obtained by tensile testing the aluminum alloy in accordance with ASTM E8 and B557 in the LT direction. [0058] Second, the stress-strain curve is enlarged so that the strain between 0% and 1.5% is visible. [0059] Third, the “YPE plateau” is determined. The YPE plateau is defined as strain from onset of significant yielding on the stress strain curve to the end of the plateau. The YPE end point is the start of persistent work-hardening. [0060] Finally, the YPE is determined by the difference of the total strain at the start and end of the plateau and expressed as a percentage with two decimal places. [0061] The YPE must be $\leq 0.60\%$ for the alloy to be classified as a reduced stretcher strain.

[0062] “Type A Lüdering” and the like means flamboyant stretcher strain marks as per Romhanji, Endre, et al. “On the Al—Mg alloy sheets for automotive application: Problems and solutions,” *Metalurgija* 10.3 (2004): 205-216, which is incorporated herein by reference in its entirety.

[0063] As used herein, “recrystallized microstructure” means a microstructure having at least 50 vol. % first type grains. In one embodiment, a recrystallized microstructure comprises at least 60 vol. % first type grains, or at least 70 vol. % first type grains, or at least 75 vol. % first type grains, or at least 80 vol. % first type grains, or at least 85 vol. % first type grains, or at least 90 vol. % first type grains, or at least 95 vol. % first type grains.

[0064] As used herein, “unrecrystallized microstructure” means a microstructure having less than 50 vol. % first type grains.

[0065] As used herein, “recovered microstructure” means the overall grain structure remains unrecrystallized while the substructure may change such as cell size growth, dislocation annihilation, and dislocation network reconstruction, among others.

vii. Microstructure Assessment Procedure

[0066] The following procedures and definitions apply to measuring microstructure features (e.g., percent recrystallization, texture) for products made in accordance with present patent application.

A. Recrystallization Determination Procedure

[0067] “Percent recrystallized” and the like means the volume percent of a wrought aluminum alloy product having recrystallized grains. The amount of recrystallized grains is determined by EBSD (electron backscatter diffraction) analysis of a suitable number of SEM micrographs of the wrought aluminum alloy product, as per this Recrystallization Determination Procedure. Generally at least 5 micrographs should be analyzed.

[0068] “Recrystallized grains” means those grains of a crystalline microstructure that meet the “first grain criteria”, defined below, and as measured using the OIM (Orientation Imaging Microscopy) sampling procedure, described below.

[0069] The OIM analysis is to be completed through the full thickness of the sheet sample on the L-ST plane, using the OIM sample procedure, below. The size of the sample to be analyzed will generally vary by gauge. Prior to measurement, the OIM samples are prepared by standard metallographic sample preparation methods. For example, the OIM samples are metallographically prepared and then polished (e.g., using 0.05 micron colloidal silica). The samples are then anodized in Barker's reagent, a diluted fluoroboric acid solution, for 90 seconds. The samples are then stripped using an aqueous phosphoric acid solution containing chromium trioxide, and then rinsed and dried.

[0070] The “OIM sample procedure” is as follows: [0071] The software used is APEX EBSD Collection Software, Version 2 (EDAX Inc., New Jersey, U.S.A.), or equivalent, which is connected to a Velocity EBSD camera (EDAX Inc., New Jersey, U.S.A.), or equivalent. The SEM is an APREO S Field Emission Gun (Thermo Fisher Scientific. Waltham, MA, U.S.A.), or equivalent. [0072] OIM run conditions are 68° tilt with a 18 mm working distance and an accelerating voltage of 20 kV with dynamic focusing and an instrument-specified beam current of 51 nA (nanoamps). The mode of collection is hexagonal grid. A selection is made such that orientations are collected in the analysis (i.e., Hough peaks information is not collected). The area

size per scan (i.e., the frame) is 2.0 mm by 1 mm for 2 mm gauge samples at 1 micron steps at 40×. Different frame sizes can be used depending upon gauge. The collected data is output in an *.osc file. This data may be used to calculate the volume fraction of first type grains, as described below. [0073] Calculation of volume fraction of first type grains: The volume fraction of first type grains is calculated using the data of the *.osc file and the OIM Analysis Software (EDAX Inc., New Jersey, U.S.A.), version 8.1.0, or equivalent. Prior to calculation, two-step data cleanup may be performed. First, for any points whose confidence index is below a threshold of 0.08, a neighbor orientation correlation clean-up is performed. Second, a grain dilation clean-up is performed for any grain smaller than 3 data points. Then, the amount of first type grains is calculated by the software using the first grain criteria (below). [0074] First grain criteria: Grain average misorientation (GAM) is calculated. All of “apply partition before calculation”, “include edge grains”, and “ignore twin boundary definitions” should be required. Any grain whose GAM is $\leq 1^\circ$ is a first type grain.

[0075] “First grain volume” (FGV) means the volume fraction of first type grains of the crystalline material.

[0076] “Percent Recrystallized” is determined via the formula: $FGV \times 100\%$.

[0077] The term “grain” has the meaning defined in ASTM E112 § 3.2.2, i.e., “the area within the confines of the original (primary) boundary observed on the two-dimensional plane of-polish or that volume enclosed by the original (primary) boundary in the three-dimensional object”.

[0078] “Grain size” is calculated by the following equation:

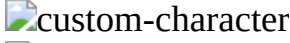
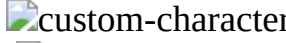
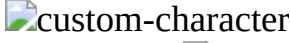
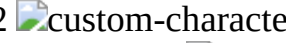
[00001] $d_i = \sqrt{4A_i}$ [0079] wherein $A_{sub.i}$ is the area of the individual grain as measured using commercial software OIM Analysis Software, version 8.1.0 or equivalent; and [0080] wherein $d_{sub.i}$ is the calculated individual grain size assuming the grain is a circle.

[0081] “Area weighted average grain size” is calculated by the following equation:

[00002] $\bar{d} = (\sum_{i=1}^n A_i d_i) / (\sum_{i=1}^n A_i)$ [0082] wherein $A_{sub.i}$ is the area of each individual grain as measured using commercial software OIM Analysis Software, version 8.1.0 or equivalent; [0083] wherein $d_{sub.i}$ is the calculated individual grain size assuming the grain is a circle; and [0084] wherein $\bar{d}$ is the area weighted average grain size.

B. Texture

[0085] “Texture” means a preferred orientation of at least some of the grains of a crystalline structure. Texture components resulting from production of aluminum alloy products may include one or more of copper, S texture, brass, cube, and Goss texture, to name a few. Each of these texture components is defined in Table A, below.

TABLE-US-00001 TABLE A Bunge Kocks Texture component Miller Indices (ϕ_1 , Φ , ϕ_2) (Ψ , Θ , Φ) copper {112}  custom-character 111  custom-character 90, 35, 45 0, 35, 45 S {123}  custom-character 634  custom-character 59, 37, 63 149, 37, 27 brass {110}  custom-character 1 12  custom-character 35, 45, 0 55, 45, 0 Cube {1 0 0} <001> 0, 0, 0 0, 0, 0 Goss {110}  custom-character 001  custom-character 0, 45, 0 0, 45, 0

[0086] EBSD data for texture quantification are that same data that are generated as described above to determine “grain size” and “percent recrystallized.” The quantification of texture components present is done by the EBSD software, i.e. OIM Analysis Software, version 8.1.0 or equivalent. The first step is to align the EBSD data from the L-ST plane into the more commonly used L-LT reference plane. Quantification of texture components present (Cube %, Goss %, Brass %, S %, Copper %) is to be determined as the number fraction of measured points assigned to a specific texture component. Points are assigned to a texture component if the misorientation angle deviates from the ideal orientation by less than 15 degrees. This number fraction is multiplied by 100 to find the percentage of each texture component in the sample.

viii. Miscellaneous

[0087] These and other aspects, advantages, and novel features of this new technology are set forth in part in the description that follows and will become apparent to those skilled in the art upon

examination of the following description and figures, or may be learned by practicing one or more embodiments of the technology provided for by the present disclosure.

[0088] Among those benefits and improvements that have been disclosed, other objects and advantages of this invention will become apparent from the following description taken in conjunction with the accompanying figures. Detailed embodiments of the present invention are disclosed herein; however, it is to be understood that the disclosed embodiments are merely illustrative of the invention that may be embodied in various forms. In addition, each of the examples given in connection with the various embodiments of the invention is intended to be illustrative, and not restrictive.

[0089] Throughout the specification and claims, the following terms take the meanings explicitly associated herein, unless the context clearly dictates otherwise. The phrases “in one embodiment” and “in some embodiments” as used herein do not necessarily refer to the same embodiment(s), though they may. Furthermore, the phrases “in another embodiment” and “in some other embodiments” as used herein do not necessarily refer to a different embodiment, although they may. Thus, various embodiments of the invention may be readily combined, without departing from the scope or spirit of the invention.

[0090] In addition, as used herein, the term “or” is an inclusive “or” operator, and is equivalent to the term “and/or,” unless the context clearly dictates otherwise. The term “based on” is not exclusive and allows for being based on additional factors not described, unless the context clearly dictates otherwise. In addition, throughout the specification, the meaning of “a,” “an,” and “the” include plural references, unless the context clearly dictates otherwise. The meaning of “in” includes “in” and “on”, unless the context clearly dictates otherwise.

[0091] While a number of embodiments of the present invention have been described, it is understood that these embodiments are illustrative only, and not restrictive, and that many modifications may become apparent to those of ordinary skill in the art. Further still, unless the context clearly requires otherwise, the various steps may be carried out in any desired order, and any applicable steps may be added and/or eliminated.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0092] FIG. 1 is a process flow diagram illustrating one embodiment of a method for processing 5xxx aluminum alloys in accordance with embodiments herein.

[0093] FIG. 2 is a process flow diagram illustrating embodiments relating to the post cold roll annealing step (300) of FIG. 1.

DETAILED DESCRIPTION

Example 1

[0094] An industrial scale ingot of the 5182 aluminum alloy was direct chill (DC) cast, then conventionally homogenized, scalped/peeled and then plant hot rolled to an intermediate gauge. Specimens of the intermediate gauge material were then processed at the lab-scale by cold roll processing and then post cold roll annealing (final annealing) as per the conditions shown in Table 1, below, thereby producing various sheet products (Products 1-5). Products 1-2 represent conventional processing to either an O temper (Product 1) or OE21 temper (Product 2). Products 3-5 represent materials made in accordance with the new processing techniques described herein, with Products 3 and 5 being particularly preferred, achieving a YPE of not greater than 0.60% (described below) and no Type A Lüdering.

TABLE-US-00002
TABLE 1 Example 1 Processing of 5182 Sheet Products Final Gauge Post Cold Roll Anneal Product Cold Roll Processing inch. (mm) (Final Anneal) 1 64% reduction to final gauge 0.052 (1.32) 675° F. for 2 hours 2 40% reduction, then an intermediate 0.065 (1.65) 675° F.

for 2 hours anneal of 675° F. for 2 hours, then 25% reduction to final gauge 3 56% reduction to final gauge 0.063 (1.60) 450° F. for 3 days and then 825° F. for 2 hours 4 56% reduction to final gauge 0.063 (1.60) 450° F. for 1 day and then 825° F. for 2 hours 5 56% reduction to final gauge 0.063 (1.60) 450° F. for 2 days and then 825° F. for 2 hours

[0095] After processing, the mechanical properties and microstructures of Products 1-5 were evaluated, the results of which are shown in Tables 2-3, below. All mechanical property values are the average of at least duplicate specimens, unless otherwise indicated. Strength and elongation properties are in the LT direction. Grain size, texture (15 degrees), dispersoid fraction, and percent recrystallization were determined in accordance with the Microstructure Assessment Procedure, described previously.

[0096] To evaluate YPE and Type A Lüdering/stretcher-strain marks, specimens of the sheet products were stretched 1% in the LT direction and then painted. Type A Lüdering was evident in Products 1 and 4. The YPE results are shown in Table 2, below.

TABLE-US-00003 TABLE 2 Mechanical Properties of Example 1 Sheet Products UTS TYS U. Elong. T. Elong. YPE Product (MPa) (MPa) (%) (%) (%) 1 271 137 20.3 30.1 1.21 2 270 116 20.4 24.8 0.40 3 266 114 23.9 26.7 0.55 4 267 119 23.9 26.9 0.69 5 267 116 24.1 27.3 0.60

TABLE-US-00004 TABLE 3 Microstructure Characteristics of Example 1 Sheet Products Area Weighted % ReX Grain Size Brass Copper S Product (GAM) (micrometers) (15 Deg) (15 Deg) (15 Deg) 1 99 14.8 12% 8% 21% 2 97 42.4 14% 7% 23% 3 98 36.5 10% 6% 16% 4 99 25.4 10% 6% 18% 5 99 29.3 9% 6% 18%

[0097] As shown, conventional 5182-O temper realizes a very poor YPE of 1.21%, which is consistent with the presence of its Type A Lüdering/stretcher-strain marks. Indeed, 5182-O temper is known to realize significant Type A Lüdering, resulting in an inability to be utilized in cosmetic applications. See Ebenberger P, Uggowitzer P J, Gerold B, Pogatscher S., *Effect of Compositional and Processing Variations in New 5182-Type AlMgMn Alloys on Mechanical Properties and Deformation Surface Quality. Materials* (Basel). 2019 May 20;12(10):1645. doi: 10.3390/ma12101645. PMID: 31137562; PMCID: PMC6566914.

[0098] Conversely, 5182-OE21 realizes low YPE (0.40%), which is consistent with the absence of Type A Lüdering, but 5182-OE21 processing requires an intermediate annealing step between cold rolling. This process is expensive and time consuming. Multiple cold rolling passes are also required, which is also expensive and time consuming. As shown herein, in some embodiments of the present disclosure, the new methods described herein may employ a single cold rolling pass. In other embodiments, the new methods described herein may employ multiple cold rolling passes.

[0099] Products 3 and 5 also had no Type A Lüdering but without any intermediate annealing step. Products 3 and 5 realize a low YPE ($\leq 0.60\%$) and strengths and elongation values consistent with 5182-OE21 temper. Products 3 and 5 have an area weighted average grain size of less than that of the 5182-OE21 temper as well as less Brass and S texture.

[0100] Product 4 had Type A Lüdering and realized a YPE slightly above the 0.60% YPE threshold, indicating more than 1 day of final annealing at 450° F. may be required to achieve a YPE of $\leq 0.60\%$ in some circumstances.

Example 2

[0101] Three industrial scale ingots of the 5182 aluminum alloy were direct chill (DC) cast, then conventionally homogenized, scalped/peeled and then plant hot rolled to an intermediate gauge. Some specimens of the intermediate gauge materials (Products 6-7) were then industrial (plant) scale cold roll processed and then post cold roll annealed (final annealing) as per the conditions shown in Table 4. Other specimens of the intermediate gauge materials (Products 8-9) were then cold roll processed at the industrial scale followed by lab-scale post cold roll annealing (final annealing) as per the conditions shown in Table 4. Products 6-7 represent conventional processing to either an O temper (Product 6) or OE21 temper (Product 7). Products 8-9 represent materials made in accordance with the new processing techniques described herein.

TABLE-US-00005 TABLE 4 Example 2 Processing of 5182 Sheet Products Final Gauge Post Cold Roll Anneal Product Cold Roll Processing inch. (mm) (Final anneal) 6 64% reduction to final gauge 0.059 (1.50) 665° F. for 3 hours 7 40% reduction, then an intermediate 0.063 (1.60) 675° F. for 2 hours anneal of 675° F. for 2 hours, then 25% reduction to final gauge 8 64% reduction to final gauge 0.059 (1.50) 450° F. for 3 days and then 825° F. for 2 hours 9 48% reduction to final gauge 0.084 (2.13) 450° F. for 3 days and then 825° F. for 2 hours

[0102] After processing, the mechanical properties and microstructures of Products 6-9 were evaluated, the results of which are shown in Tables 5-6, below. All mechanical property values are the average of at least duplicate specimens, unless otherwise indicated. Strength and elongation properties are in the L direction, except the OE21 material properties are relative to the LT direction. Grain size, texture, dispersoid fraction, and percent recrystallization were determined in accordance with the Microstructure Assessment Procedure, described previously.

[0103] To evaluate YPE and Type A Lüdering/stretcher-strain marks, specimens of the alloys were stretched 1% and then painted. Type A Lüdering/stretcher-strain marks were evident only in Product 6. The YPE results are shown in Table 5, below.

TABLE-US-00006 TABLE 5 Mechanical Properties of Example 2 Sheet Products TYS UTS U. Elong. T. Elong. YPE Product (MPa) (MPa) (%) (%) (%) 6 144 293 19.8 25.2 0.93 7 117 276 20.0 25.0 0.16 8 123 281 20.5 24.9 0.52 9 116 281 19.7 26.3 0.16

TABLE-US-00007 TABLE 6 Microstructure Characteristics of Example 2 Sheet Products Area Weighted % ReX Grain Size Brass Copper S Product (GAM) (micrometers) (15 Deg) (15 Deg) (15 Deg) 6 99 16.0 9% 9% 17% 7 89 41.7 15% 7% 20% 8 99 28.6 6% 5% 13% 9 98 36.5 8% 6% 15%

[0104] Consistent with Example 1, the 5182-O temper sheet product (Product 6) realized very poor YPE at 0.93% while the 5182-OE21 (Product 7) realized low YPE at 0.16%. Products 8-9 also realized low YPE ($\leq 0.60\%$) but without an intermediate anneal. Product 8 with two cold rolling passes had higher YPE whereas Product 9 with a single cold rolling pass realized much lower YPE. The grain sizes of Products 8-9 are also lower than 5182-OE21 (Product 7) as were the Brass and S texture components.

Example 3

[0105] An industrial scale ingot of a 5182 aluminum alloy was direct chill (DC) cast, then conventionally homogenized, scalped/peeled and then hot rolled to an intermediate gauge. The intermediate gauge material was then cold rolled with a 64% reduction to a final gauge of 0.059 inch (1.50 mm). No intermediate anneal was completed in between hot rolling and cold rolling, or during cold rolling. After cold rolling, the final gauge material was annealed at various conditions, as shown in Table 7, below. The mechanical properties of the sheet products were then tested, the results of which are also shown in Table 7 below. To evaluate YPE, specimens of the alloys were stretched 1%. The YPE results are also shown in Table 7, below.

TABLE-US-00008 TABLE 7 Processing and Results of Example 3 Sheet Products Post Cold Roll Anneal UTS TYS U. Elong. T. Elong. YPE Product (Final Anneal) (MPa) (MPa) (%) (%) (%) 10 450° F. for 24 hours and 275 128 25.4 27.1 0.86 then 825° F. for 2 hours 11 450° F. for 48 hours and 281 123 19.9 26.5 0.46 then 825° F. for 2 hours 12 450° F. for 72 hours and 281 123 23.2 26.8 0.54 then 825° F. for 2 hours 13 450° F. for 3 hours and 278 139 22.1 25.3 1.19 then 825° F. for 2 hours 14 825° F. for 5 days (120 276 134 25.8 28.1 1.17 hours) 15 450° F. for 96 hours and 274 119 23.3 26.3 0.56 then 825° F. for 6 hours 16 500° F. for 24 hours and 279 137 23.9 27.9 0.90 then 825° F. for 4 hours

[0106] As shown, a two-step final anneal is beneficial over a single-step final anneal. For instance, Product 14 with a single-step final anneal over 5 days realized a YPE of 1.17%, which is significantly higher than that of the two-step final annealed products annealed for at least 24 hours at 450° F. or 500° F. Similarly, a short first anneal step may be insufficient. For instance, Products 10 and 13 with first step anneal times of 24 hours and 3 hours, respectively, were unable to achieve

a YPE of $\leq 0.60\%$. Conversely, Products 11-12 and 15 with first step anneal times of 48, 72 and 96 hours, respectively, achieved YPEs below 0.60%. Higher first step anneal temperatures may also be insufficient to achieve a low YPE. For instance, Product 16 with a first step anneal of 500° F. for 24 hours realized similar or worse YPE than Product 10 with a first step anneal of 450° F. for 24 hours.

[0107] While various embodiments of the present disclosure have been described in detail, it is apparent that modifications and adaptations of those embodiments will occur to those skilled in the art. However, it is to be expressly understood that such modifications and adaptations are within the spirit and scope of the present disclosure.

Claims

1. A method comprising: (a) hot rolling a 5xxx aluminum alloy to an intermediate gauge product; (b) after the hot rolling step, conducting a cold rolling operation to achieve a final gauge product, wherein the cold rolling operation includes at least one cold rolling step, and wherein no intermediate annealing occurs prior to or during the cold rolling operation; (c) after the cold rolling operation, post cold roll annealing the final gauge product, wherein the post cold roll annealing step (c) comprises: (i) first annealing at a first temperature for a first duration, wherein, due to the first annealing step, the final gauge product contains a recovered but unrecrystallized microstructure; and (ii) after the first annealing step, applying a second annealing step to the final gauge product, thereby converting at least some of the recovered but unrecrystallized microstructure to a recrystallized microstructure.
2. The method of claim 1, wherein the post cold roll annealing step comprises annealing the final gauge product at a temperature and for a time sufficient for the final gauge product to realize a YPE of not greater than 0.60%.
3. The method of claim 2, wherein the final gauge product realizes at least one of: (A) an area weighted average grain size of not greater than 40 micrometers; (B) a Brass texture amount of not greater than 14%; and (C) an S texture amount of not greater than 19%.
4. The method of claim 3, wherein, due to the second annealing step, the final gauge product realizes at least two of (A), (B) and (C).
5. The method of claim 1, wherein the final gauge product realizes an area weighted average grain size of from 15 to 39 micrometers.
6. The method of claim 5, wherein the final gauge product realizes a Brass texture amount of from 1% to 13%.
7. The method of claim 6, wherein the final gauge product realizes an S texture amount of from 5% to 18%, or an S texture amount of not greater than 17%, or an S texture amount of not greater than 16%, or an S texture amount of not greater than 15%, or an S texture amount of not greater than 14%.
8. The method of claim 1, wherein the second temperature is at least 25° F. higher than the first temperature; wherein the first duration is at least 30 minutes; wherein the second duration is at least 1 second.
9. The method of claim 8, wherein the first temperature is below a recrystallization temperature of the final gauge product, and wherein the second temperature is above the recrystallization temperature of the final gauge product.
10. The method of claim 9, wherein the first temperature is from 350° F. to 500° F.
11. The method of claim 10, wherein the first duration is from at least 1 hour to not greater than 10 days.
12. The method of claim 11, wherein the second temperature is from greater than 500° F. to 900° F.
13. The method of claim 12, wherein the second duration is from 5 seconds to 48 hours.
14. The method of claim 1, wherein the cold rolling operation is accomplished with a single cold

rolling pass.

15. The method of claim 1, wherein the cold rolling operation is accomplished with multiple cold rolling passes.

16. The method of claim 1, wherein the cold rolling operation comprises reducing a thickness of the intermediate gauge product by from 25% to 80%.

17. The method of claim 1, wherein the hot rolling comprises hot rolling an ingot or a continuously cast strip of the 5xxx aluminum alloy.

18. A 5xxx aluminum alloy sheet product comprising from 4.0 to 6.0 wt. % Mg and having a thickness of from 0.5 to 5.0 mm, wherein the 5xxx aluminum alloy sheet product is at least 60% recrystallized and realizes an YPE of not greater than 0.60%; wherein the 5xxx aluminum alloy sheet product realizes at least one of: (A) an area weighted average grain size of not greater than 40 micrometers; (B) a Brass texture amount of not greater than 14%; and (C) an S texture amount of not greater than 19%.

19. The 5xxx aluminum alloy sheet product of claim 18, comprising: from 4.0 to 5.0 wt. % Mg up to 0.50 wt. % Mn; up to 0.75 wt. % Zn; up to 0.35 wt. % Fe; up to 0.20 wt. % Si; up to 0.15 wt. % Cu; up to 0.10 wt. % Cr; up to 0.10 wt. % Ti; the balance being aluminum, optional incidental elements and impurities.

20. The 5xxx aluminum alloy sheet product of claim 19, wherein the 5xxx aluminum alloy includes at least 0.25 wt. % Zn, or at least 0.30 wt. % Zn, or at least 0.35 wt. % Zn, or at least 0.40 wt. % Zn, or at least 0.45 wt. % Zn, or at least 0.50 wt. % Zn.
